

ConstructionAR Platform Requirements Document

1. Document Purpose

This document defines the product requirements for the proposed ConstructionAR Platform based on the client's stated vision. Its purpose is to translate the vision into a structured set of requirements that can guide product planning, technical architecture, implementation sequencing, and stakeholder alignment.

This document intentionally preserves the client's broader platform vision while distinguishing between:

- MVP requirements required for the first meaningful product release
- Expansion requirements intended for later phases

Where tradeoffs are necessary, this document favors fidelity to the client's stated goals while clearly identifying scope boundaries needed for practical delivery.

2. Product Vision

ConstructionAR is an augmented reality application intended for both mobile phone/tablet and headset devices, including platforms such as Meta devices and Apple Vision Pro class hardware. The platform will scan real-world spaces and create interactive digital models that allow users to place virtual construction components, building-related objects, and realistic product representations into the environment in real time.

The long-term goal is to provide a unified workflow that combines:

- Space scanning
- Blueprint import and reference
- Blueprint generation
- 3D modeling
- AR object and product placement
- Measurement tools
- Rule-based conflict detection
- Construction coordination and field validation
- Cross-device interaction
- Project photos and field notes
- Collaboration
- Reporting
- Project documentation

The platform is intended to support both residential and commercial use cases, including immersive visualization for homeowners and customers as well as coordination, validation, and documentation workflows for contractors, architects, engineers, and construction managers.

For residential smart-home applications, the long-term vision extends beyond basic object placement. ConstructionAR should allow customers and professionals to visualize realistic, correctly scaled representations of actual products in the customer's physical environment and, in later phases, simulate certain aspects of how the completed smart-home system could function.

For professional construction applications, the long-term vision includes combining real-world spatial information with project plans, proposed installations, multiple building trades, and actual field conditions to support coordination and validation before and during construction.

3. Business Context

Construction, renovation, and building-systems projects are often planned using disconnected tools that separate visualization, design, coordination, validation, and documentation. This creates friction in both sales and execution:

- Clients struggle to visualize proposed work in their actual space.
- Contractors and designers make placement decisions with limited spatial context.
- Trade conflicts and clearance issues are often identified late.
- Documentation workflows are separated from field visualization activities.
- Different stakeholders rely on different tools, which slows decision-making and increases rework risk.

ConstructionAR is intended to reduce those gaps by using the real environment as the primary interface for design review, planning, and validation.

4. Product Scope

4.1 In-Scope Product Direction

The product direction defined by the client includes the following major capability areas:

- Real-world space scanning
- Creation of interactive digital spatial models
- Import and reference of existing floor plans/blueprints
- Placement of virtual construction and building-system objects
- Placement of realistic, correctly scaled product representations
- Visualization of layouts in the actual space
- Measurement and fit-validation workflows
- Rule-based conflict and clash detection
- Construction coordination and field validation
- Cross-device phone/tablet/headset interaction
- Project photos and field notes
- Collaboration and review workflows
- Reporting and project documentation

4.2 MVP Scope

The MVP shall focus on the smallest end-to-end workflow that demonstrates real user value and technical feasibility.

The MVP should include:

- Scanning a real-world space
- Creating a usable spatial scene representation
- Importing an existing 2D floor plan/blueprint and using it as a project reference

- Browsing a limited library of virtual objects
- Placing and repositioning those objects within the scanned environment
- Saving and reopening a project or layout
- Supporting basic measurement-aware visualization
- Capturing basic project/site photos and notes
- Producing simple visual outputs for sharing or review

4.3 Post-MVP Scope

The following areas are part of the broader platform vision but are not required for the initial MVP unless later directed:

- Advanced clash detection across multiple trades
- Advanced construction coordination and field validation
- Real-time multi-user collaboration
- Full phone/tablet/headset synchronization
- Advanced smart-home product simulation
- Manufacturer/product integrations
- Automated blueprint generation
- Deep reporting and documentation pipelines
- Extensive CAD/BIM interoperability
- Enterprise administration and large-scale project governance features

5. Stakeholders and User Groups

5.1 Primary Stakeholders

- Client sponsor
- Product owner or business lead
- Technical architecture and engineering team
- Pilot customers and early adopters

5.2 User Groups

Residential Users

- Homeowners
- Smart home integrators
- Residential contractors
- Interior designers

Commercial Users

- General contractors
- Electrical contractors
- HVAC contractors

- Plumbing contractors
- Fire protection contractors
- Fire alarm contractors
- Low-voltage contractors
- Architects
- Engineers
- Construction managers
- Facility managers

6. Supported Platforms and Devices

6.1 Target Platforms

The product shall be designed with a cross-platform strategy that supports:

- Mobile phones and tablets with AR capabilities
- Meta headset devices
- Apple Vision Pro class devices

The architecture should also preserve the ability for mobile/tablet and headset devices to operate together within a shared ConstructionAR session.

Phone/Tablet = Technician/Controller



Shared ConstructionAR Session



Headset = Immersive Visualization/Interaction Device

Changes made on one participating device should eventually be capable of appearing on another participating device.

6.2 MVP Platform Constraint

The MVP shall launch with a limited and explicitly defined support matrix. Full functional parity across all target devices is not required for the first release.

Full cross-device synchronization is not required for the initial POC/MVP, but the architecture should account for its future implementation.

7. User Goals

The system should enable users to:

- Understand a real space digitally without relying only on static plans or rough measurements
- Visualize proposed installations before work begins
- Compare placement options in context
- Evaluate whether selected materials, devices, or equipment fit the environment
- Present design concepts more clearly to customers and stakeholders
- Reduce avoidable field conflicts and installation errors
- Document relevant project/site conditions

- Compare planned installations against actual field conditions in later phases

8. Functional Requirements

8.1 Spatial Capture

FR-1

The system shall allow a user to scan a real-world environment using a supported device.

FR-2

The system shall convert the captured environment into a spatial representation that can be used for object placement and visualization.

FR-3

The system shall retain enough environmental understanding to support anchored placement of virtual objects within the scanned space.

FR-4

The system should allow a user to rescan or refresh the environment when the original scan is incomplete or inaccurate.

8.2 Project and Scene Management

FR-5

The system shall allow a user to create a new project or scene.

FR-6

The system shall allow a user to save a project containing spatial context and placed virtual objects.

FR-7

The system shall allow a user to reopen a previously saved project.

FR-8

The system should allow a user to duplicate or version a project as design alternatives are explored.

8.3 Object Library and Asset Access

FR-9

The system shall provide access to a limited but organized library of virtual objects for the MVP.

FR-10

The object library should support categorization by use case or trade, such as smart home, electrical, HVAC, plumbing, fire protection, fire alarm, low voltage, furnishings, or general building components.

FR-11

Each object should include enough metadata to support basic identification, sizing, and placement. Where appropriate, the platform should support realistic, correctly scaled representations of actual products rather than only generic object representations.

8.4 AR Object Placement and Editing

FR-12

The system shall allow a user to place a virtual object into the scanned environment.

FR-13

The system shall anchor placed objects relative to the scanned environment.

FR-14

The system shall allow a user to move, rotate, and reposition a placed object.

FR-15

The system should allow a user to remove placed objects from the scene.

FR-16

The system should support placement of multiple objects within a single project.

8.5 Visualization

FR-17

The system shall render placed virtual objects in context within the scanned environment.

FR-18

The system should support an immersive visualization experience appropriate to the target device form factor.

FR-19

The system should preserve scale relationships sufficient for layout review and pre-installation visualization.

FR-19A - Residential Product Visualization

For applicable residential smart-home workflows, the platform should support realistic, correctly scaled representations of products so users can evaluate appearance, size, location, and alternative product or placement options within the actual space. Post-MVP releases may support simulated behavior of selected smart-home devices and systems, such as lighting scenes, motorized shades, touchscreen interfaces, camera viewing areas, displays, audio zones, and multi-device automation scenes. The purpose is not necessarily to fully emulate manufacturer hardware, but to allow customers to understand both what a proposed installation could look like and how it could function.

8.6 Measurement and Validation

FR-20

The system should provide basic measurement-aware placement support for the MVP.

FR-21

The system should allow a user to evaluate whether an object appears to fit within the available space.

FR-22

The system may provide simple distance, spacing, or clearance cues in the MVP where technically feasible.

8.7 Conflict Detection, Construction Coordination, and Field Validation

FR-23

The platform shall be architected to support rule-based conflict detection, construction coordination, and field validation in later phases.

FR-24

Post-MVP releases should support clash or clearance validation for prioritized object types and trade workflows.

The long-term construction coordination capability should support combining:

- Spatial scans of the actual jobsite

- Digital plans, blueprints, and/or 3D models
- Proposed component and equipment locations
- Multiple construction trades and building systems
- Measurements, required clearances, and spatial relationships
- Actual field conditions

The goal is to allow project stakeholders to visualize proposed installations within the actual jobsite, compare planned work against real-world conditions, and identify potential issues involving placement, routing, accessibility, clearances, or interference between trades before installation occurs.

FR-25

Advanced multi-trade conflict detection is not required for the MVP. Post-MVP releases should also support field validation workflows that allow users to compare planned work against actual field conditions or installed work and document relevant discrepancies or changes.

8.8 Collaboration and Review

FR-26

The MVP shall allow users to create simple shareable outputs, such as screenshots or visual exports, for review.

FR-27

Post-MVP releases should support structured review, annotations, comments, or approval-oriented collaboration workflows.

FR-28

Real-time multi-user collaborative editing is not required for the MVP. The architecture should preserve future shared-session interaction between supported mobile/tablet and headset devices.

8.9 Reporting and Documentation

FR-29

The MVP should support simple output generation sufficient to communicate a proposed layout or design state.

FR-30

The MVP should support basic project/site documentation, including photo capture, field notes, and notes associated with the overall project and, where practical, specific objects or locations. Later phases should support richer project reports, documentation, and traceable project artifacts.

FR-31

Automated blueprint generation is not required for the MVP.

8.10 Blueprint Import and Reference

FR-32

The MVP should allow a user to import an existing 2D floor plan or blueprint and associate it with a ConstructionAR project.

FR-33

The imported floor plan or blueprint should be available as a reference within the project. Basic blueprint import is distinct from automated blueprint generation and extensive CAD/BIM interoperability, which may remain post-MVP capabilities.

9. Non-Functional Requirements

9.1 Performance

NFR-1

The system shall provide responsive interaction during scanning, placement, and object manipulation on supported MVP devices.

NFR-2

The system should maintain stable rendering and acceptable frame performance for immersive review workflows.

9.2 Reliability

NFR-3

The system shall preserve saved projects without data loss during normal use.

NFR-4

The system should degrade gracefully when environmental understanding is weak or incomplete.

9.3 Usability

NFR-5

The core scan-to-place workflow shall be understandable to pilot users without requiring extensive operator intervention.

NFR-6

The product should be usable by non-technical stakeholders in customer-facing visualization scenarios.

9.4 Accuracy and Trust

NFR-7

The system shall communicate that the MVP supports planning and visualization workflows and is not automatically guaranteed to provide survey-grade accuracy.

NFR-8

The product should provide consistent spatial behavior sufficient to build user confidence in placement and review workflows.

9.5 Maintainability and Extensibility

NFR-9

The system architecture shall support phased expansion into clash detection, construction coordination, field validation, collaboration, reporting, smart-home simulation, and broader device coverage.

NFR-10

The product should be modular enough to allow expansion of object libraries, rule sets, output formats, product integrations, and cross-device workflows over time. The MVP architecture should not unnecessarily prevent future implementation of phone/tablet/headset synchronization, advanced smart-home visualization, multi-trade coordination, or plan/model integration.

9.6 Security and Data Handling

NFR-11

If project data or scanned environments are stored, the system shall define clear handling rules for project persistence and access control in later architecture work.

NFR-12

Enterprise-grade security and governance requirements are deferred until deployment, customer, and hosting expectations are defined.

10. MVP Use Cases

UC-1 Residential Visualization

A homeowner or integrator scans a room, selects a realistic and correctly scaled smart-home device, places it in context, compares placement options, and reviews the proposed installation before purchase or installation. Advanced simulation of device or smart-home system behavior is not required for the MVP.

UC-2 Contractor Layout Review

A contractor scans a space, references an imported 2D floor plan or blueprint, places a limited set of virtual components, and reviews approximate positioning and fit before field installation.

UC-3 Customer Presentation

A project stakeholder creates a spatial layout and shares visual outputs with a customer or internal decision-maker to accelerate approval.

UC-4 Field Documentation

A user captures project/site photos and notes so relevant field information can remain associated with the ConstructionAR project.

11. Assumptions

- The client's product vision remains the guiding source for scope and future direction.
- The first release will prioritize a narrow, high-value workflow rather than the full platform.
- Device sensor quality and AR platform capabilities will vary by hardware class.
- Precision expectations will need to be managed carefully, especially for construction workflows.
- Rich documentation, automation, integration, coordination, and simulation capabilities will require later-phase implementation.
- Mobile/tablet and headset devices may eventually participate together in shared ConstructionAR sessions.

12. Constraints

- Cross-platform AR parity may not be achievable in the MVP.
- Some advanced features depend on SDK capabilities that differ by vendor ecosystem.
- The breadth of target industries and object categories exceeds what can be modeled in the first release.
- Time and implementation risk require progressive delivery rather than full-platform delivery at launch.
- Cross-device synchronization capabilities may differ across supported device ecosystems.

13. Out of Scope for MVP

- Full real-time collaboration across multiple users
- Full phone/tablet/headset synchronization
- Universal trade support and exhaustive object libraries
- Advanced smart-home device/system simulation
- Production-grade automated blueprint generation
- Full BIM and CAD interoperability
- Advanced multi-trade construction coordination
- Comprehensive rule engines for every discipline
- Enterprise workflow management beyond core visualization and layout use cases
- Guaranteed survey-grade or engineering-grade measurement precision

14. Risks and Dependencies

14.1 Key Risks

- AR tracking, anchoring, and scene understanding may behave inconsistently across environments.
- Users may expect precision beyond what the initial hardware and software stack can reliably provide.
- The product vision spans both immersive sales workflows and professional construction coordination, which may create prioritization tension.
- Expanding too many trades or features in the MVP may create delivery risk and dilute user value.
- Cross-device functionality may vary based on device/platform capabilities.

14.2 Key Dependencies

- Final device strategy and platform selection
- Availability and suitability of AR and spatial computing SDKs
- Definition of the initial object library
- Availability of realistic product assets where required
- Alignment on the first target persona and use case
- Architecture decisions for storage, project persistence, and future extensibility
- Future cross-device communication and synchronization strategy

15. Success Criteria

The MVP will be considered successful if:

- Users can reliably scan a representative environment on supported devices.
- Users can place and reposition virtual objects within the environment with stable enough alignment for planning and visualization.
- Users can import and reference a basic 2D floor plan/blueprint within a project.
- Users can capture basic project/site photos and notes.
- Pilot users can complete a full scan-to-layout workflow with limited guidance.

- The product materially improves understanding of proposed installations versus static explanation alone.
- The MVP establishes a credible platform foundation for future validation, collaboration, smart-home simulation, cross-device interaction, reporting, construction coordination, and documentation phases.

16. Open Decision Points

The following items should be treated as active decision points rather than assumed answers:

- Which device family will be the primary MVP target
- Which user persona is the first commercial focus
- Which object categories are required for the initial release
- What level of measurement support is mandatory versus optional in the MVP
- Whether the first release is positioned more as a visualization tool, a coordination tool, or a hybrid
- What level of basic blueprint import/reference functionality is appropriate for the MVP
- Whether any phone/tablet/headset synchronization should be demonstrated in the POC/MVP or reserved for a later phase

17. Recommended Next Artifacts

The following documents should be developed next to continue the project responsibly while staying aligned with the client's vision:

- Product requirements traceability matrix
- MVP feature breakdown
- System architecture document
- User workflow diagrams
- Milestone and release plan
- Risk register